Principles of Uncertainty

Homework 1

Timon Bubnič (63220035) · October 24, 2026

Problem 2.2 (Textbook, ch 2)

Solution. Let $\mathcal{C}$ be a collection of subsets of Ω. Then $\mathcal{C} = \sigma(\mathcal{C})$ if and only if $\mathcal{C}$ is a σ-algebra.

($\Rightarrow$) $\sigma(\mathcal{C})$ is a σ-algebra by definition.

($\Leftarrow$) We always have $\mathcal{C} \subseteq \sigma(\mathcal{C})$. Since $\mathcal{C}$ is a σ-algebra containing $\mathcal{C}$, and $\sigma(\mathcal{C})$ is the smallest such, $\sigma(\mathcal{C}) \subseteq \mathcal{C}$. □

Problem 2.3 (Textbook, ch 2)

Solution. Suppose $C \subseteq D$. Let

$$\chi = \{\text{all } \sigma\text{-algebras containing } C\}, \qquad \delta = \{\text{all } \sigma\text{-algebras containing } D\}.$$

Then $\delta \subseteq \chi$: every σ-algebra that contains D also contains C, since $C \subseteq D$ (the reverse doesn't hold in general). An intersection over a larger family is smaller, so

$$\sigma(C) = \bigcap \chi \subseteq \bigcap \delta = \sigma(D).$$

□